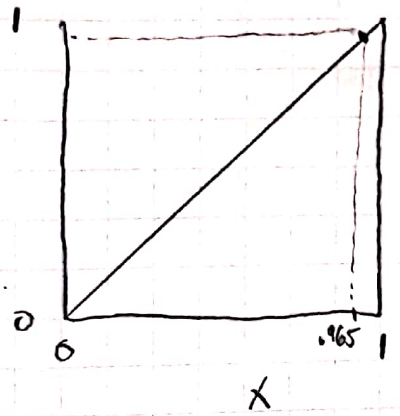
Hw # 1 - David Alvarado

8.5

1) Salt water, salt non-volatile

McCabe - Thiele diagram 3.5 wt% salt → (Plot more volatile $\frac{y}{x}$)

$100 - 3.5 = 96.5$ wt%



operating line?
Equilibrium line?

2) DePriester Chart → Allign Temp & Pressure (Ruler) ⇒ Get k values

a. 100°C & 200 kPa $k_{n\text{-hexane}} = 1.2$

b. As P ↑ k values decrease

c. As MW ↑ k values decrease

d. At what P does pre propane boil at temp of $T = -30°C$

Boil [vapor / Liquid] $k_i = \frac{y_i}{x_i}$ $k_i = 1$ $P = 170$ kPa

3) Methanol - water

a. $y = 0.60$ Plotting data in excel & finding x-value corresponding to $y = 0.60$

↳ $x = 0.22$

b. Azeotrope $x = y$; Plotting $y = x$ in the graph & seeing if it intercepts graph.
⇒ No interception ⇒ No Azeotrope

c. $X_w = 0.35$ → $\Sigma x = 1$ → $X_E = 1 - X_w = 0.65$
From graph w/ $X = 0.65$ ⇒ $y = 0.84$ ✗

~ David Alvarado

d. $X_E = 0.20$ From (data) graph $X_E = 0.20 \Rightarrow y_E = 0.579$

$X_W = 1 - X_E = 0.80$ $y_W = 1 - y_E = 0.421$

$k_i = \frac{y_i}{x_i}$ $k_E = \frac{y_E}{x_E} = \frac{0.579}{0.20}$ $k_W = \frac{y_W}{x_W} = \frac{0.421}{0.8}$

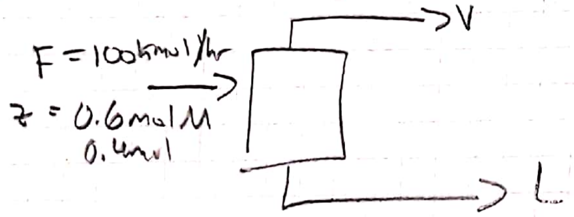
$k_M = 2.895$ $k_W = 0.52625$

e. $\alpha_{AB} = ?$ $X_E = 0.20$

$\alpha_{AB} = \frac{k_A}{k_B} \Rightarrow \alpha_{M-W} = \frac{k_M}{k_W} = \frac{2.895}{0.52625} = 5.50$ (From d.)

(11) M-W → 1.0 atm pressure

a. Feed is 60.0 mol% M $F = 100\ kmol/h$
40% of the feed is vaporized



[40% vap. → $V = 0.4 \cdot F = 40$
[60% Not → $L = 0.6\ F = 60$

eqn. $y = -\frac{L}{V} x + \frac{F}{V} z$

→ $y = -\frac{60}{40} x + \frac{100}{40}(0.6)$

$y = -\frac{3}{2} x + \frac{3}{2}$ X_M and Y_M?

b. $F = 1500\ kmol/hr$

$V = 0.4F = 0.4(1500) = 600$
$L = 0.6F = 0.6(1500) = 900$

eqn: $y = -\frac{L}{V} x + \frac{F}{V} z$

$y = -\frac{900}{600} x + \frac{1500}{600}(0.6)$

$y = -3/2\, x + 3/2$

X_M and Y_M?

c. $F = 1000\ lbmol/hr$

$z = 0.3$

$L \sim x = .2\ M$

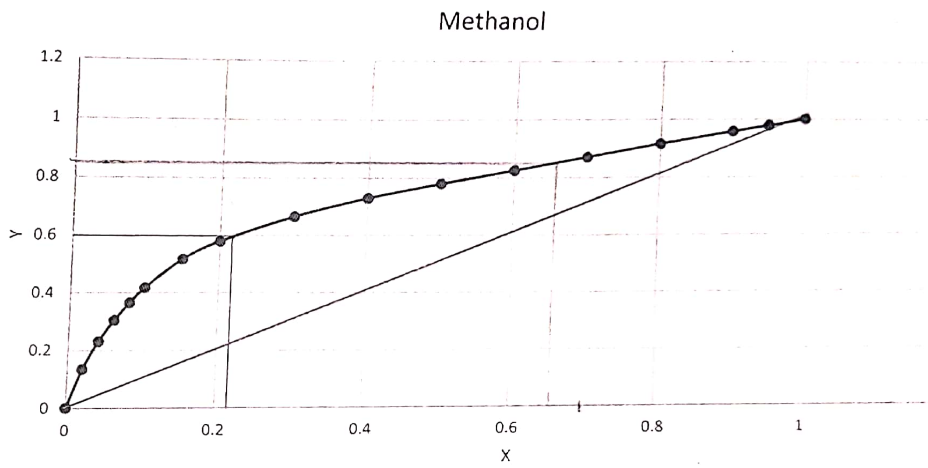
$V/F = ?$

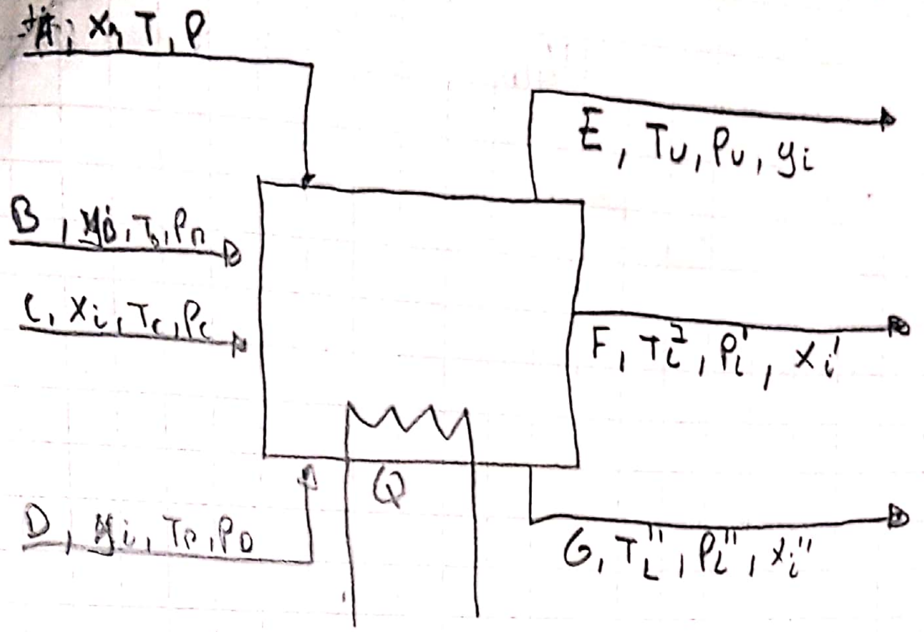
d. $L \sim 45\%$ mol Meth

$L = 1500.00\ kmol/h$

$V/F = 0.2$

Methanol liquid	Methanol Vapor
0	0
0.02	0.134
0.04	0.23
0.06	0.304
0.08	0.365
0.1	0.418
0.15	0.517
0.2	0.579
0.3	0.665
0.4	0.729
0.5	0.779
0.6	0.825
0.7	0.87
0.8	0.915
0.9	0.958
0.95	0.979
1	1





a. List + count vars.

7 flows	A, B, C, D, E, F, G	7
1 Heat	Q	1
T + P for each stream	$T_A, T_B, T_C, T_D, T_V, T_L', T_L''$	7
	$P_A, P_B, P_C, P_D, P_V, P_L', P_L''$	7
n components in 7 streams	x_i in 7 streams	7(n)
		22 + 7n

b. Write + count equations

- Total Mass Balance: $A + B + C + D = E + F + G$ — $\cancel{1}$
- Component Balances $i = 1$ to n: $Ax_{Ai} + By_{iB} + Cx_{Ci} + Dy_{Di} = Ey_i + Fx_i' + Gx_i''$ — n
- Comp. of streams $\sum_1^n x_i = 1$ — 7
- Energy Balance: — 1
- Equilibrium: $K_i' = y_i / x_i'$, $K_i'' = y_i / x_i''$ — 2n
- P: $P'_{drum} = P(x_i', T_{drum})$, $P''_{drum} = P(x_i'', T_{drum})$ — 2
- T: $T'_{drum} = T(x_i^{I}, P_{drum})$, $T''_{drum} \; T(x_i^{II}, P_{drum})$ — 2

12 + 3n

[illegible] Alvarado - Hw #3

c.

DOF = # vars. - # eqs.

$(22+7n) - (12+3n)$

$DOF = 10 + 4n$

$$\frac{dy}{dx} = f(x)$$